13

Electromagnetic Induction

13.3

Induced Currents and Magnetic Forces

13.3.1

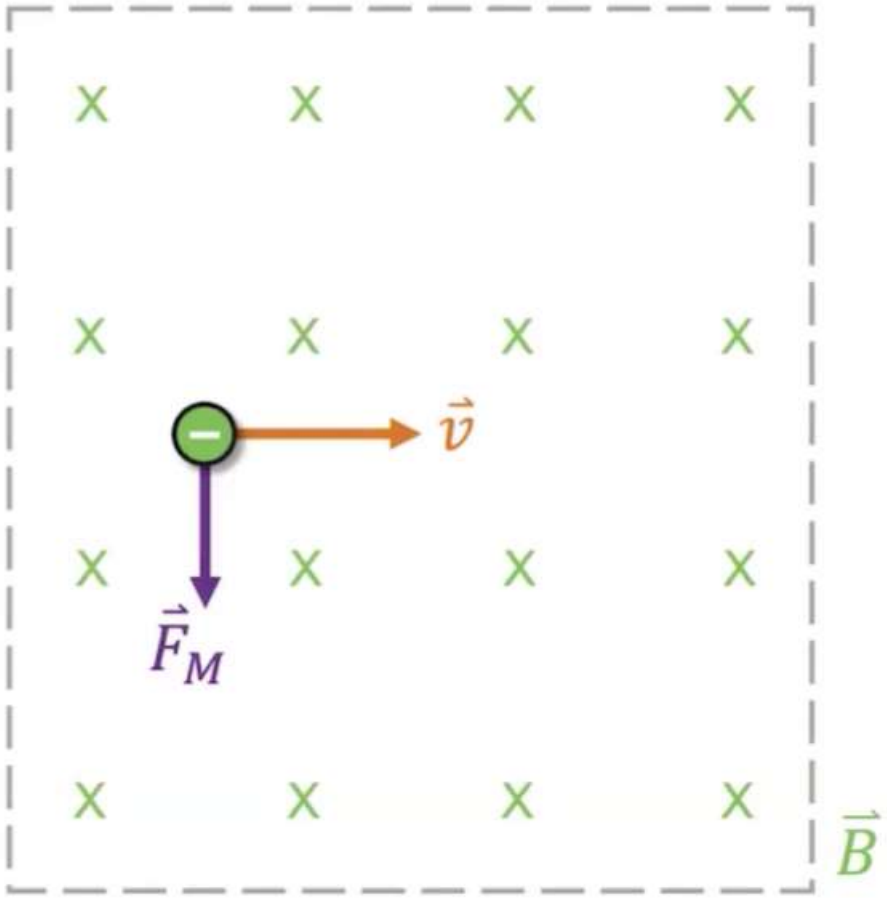
The force exerted an external magnetic field on a conducting loop with an induced current

https://www.aol.com/articles/bryson-dechambeau-brings-3d-printed-121454892.html

Magnetic Forces on Charges

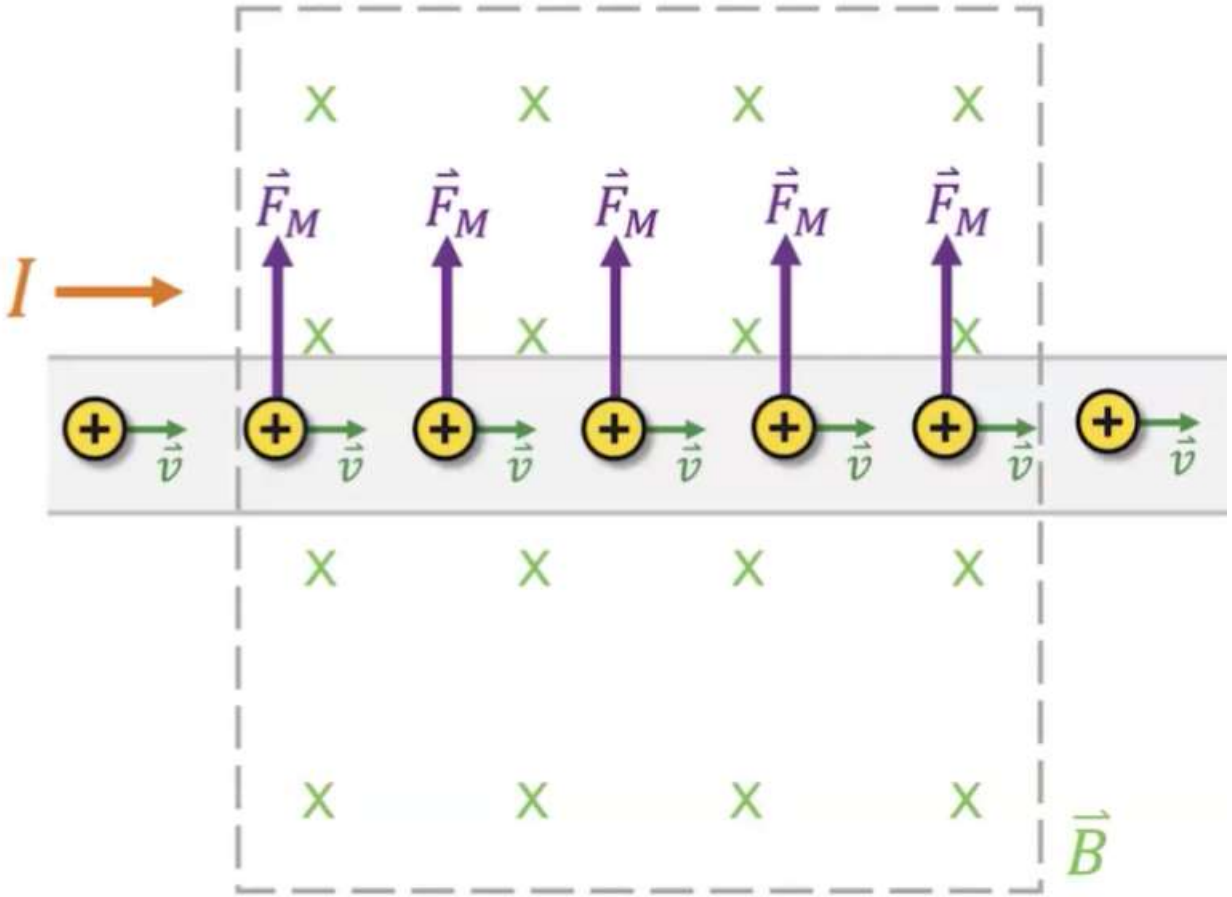
A magnetic field exerts a force on charges moving through a field.

$$\vec{F}_M = q\vec{v}\times\vec{B}$$



Magnetic Forces on Current-Carrying Wires

If the charges are inside a wire carrying a current that is inside a magnetic field, they will also experience a force from the magnetic field.

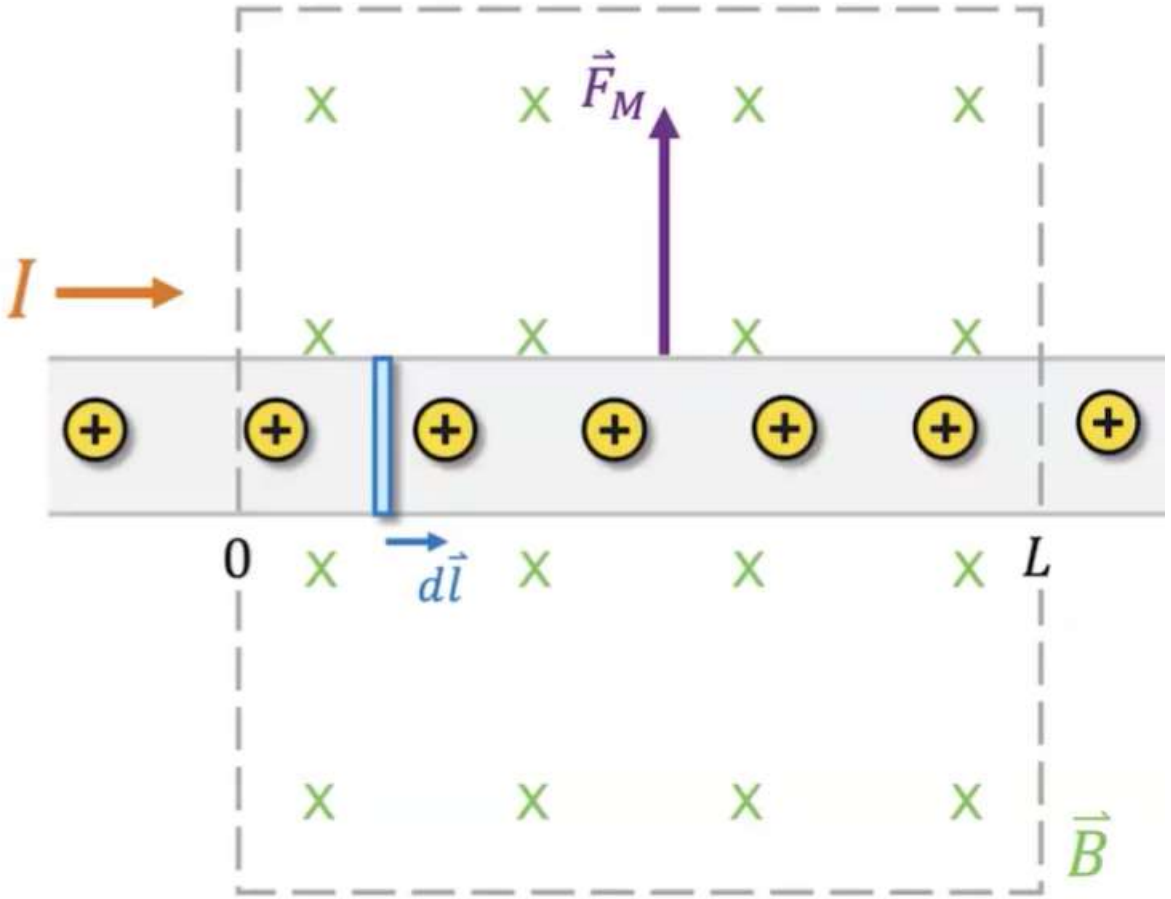


Magnetic Forces on Current-Carrying Wires

If the charges are inside a wire carrying a current that is inside a magnetic field, they will also experience a force from the magnetic field.

The sum of these can be represented by a single magnetic force on that wire.

$$\vec{F}_M = \int_0^L I(d\vec{l} \times \vec{B})$$

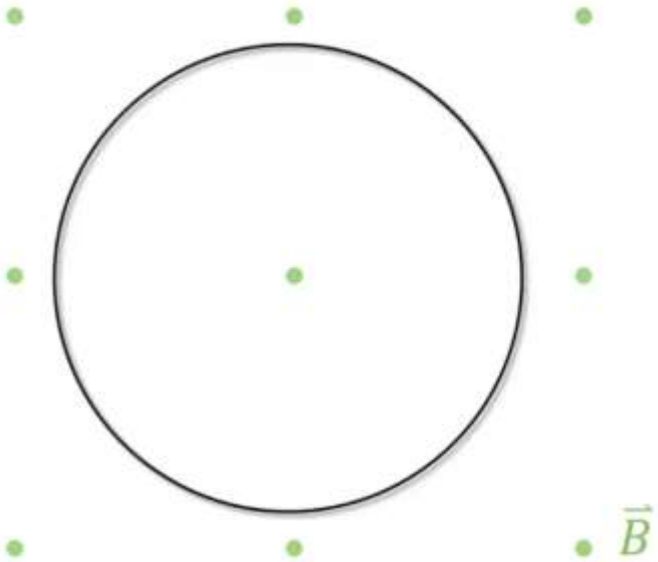


Induced emf Produces an Induced Current

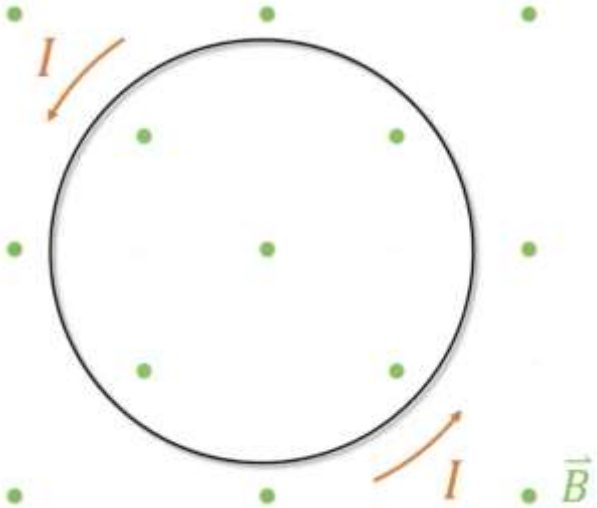
By Faraday's Law, a changing magnetic flux induces an emf in a conducting loop.

$$|\mathcal{E}| = N\left|\frac{d\Phi}{dt}\right|$$

By Lenz's Law, the current will be in a direction that opposes the change in magnetic flux.

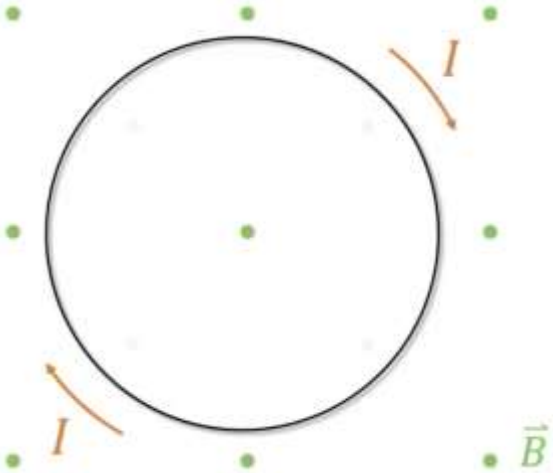


Increasing Flux



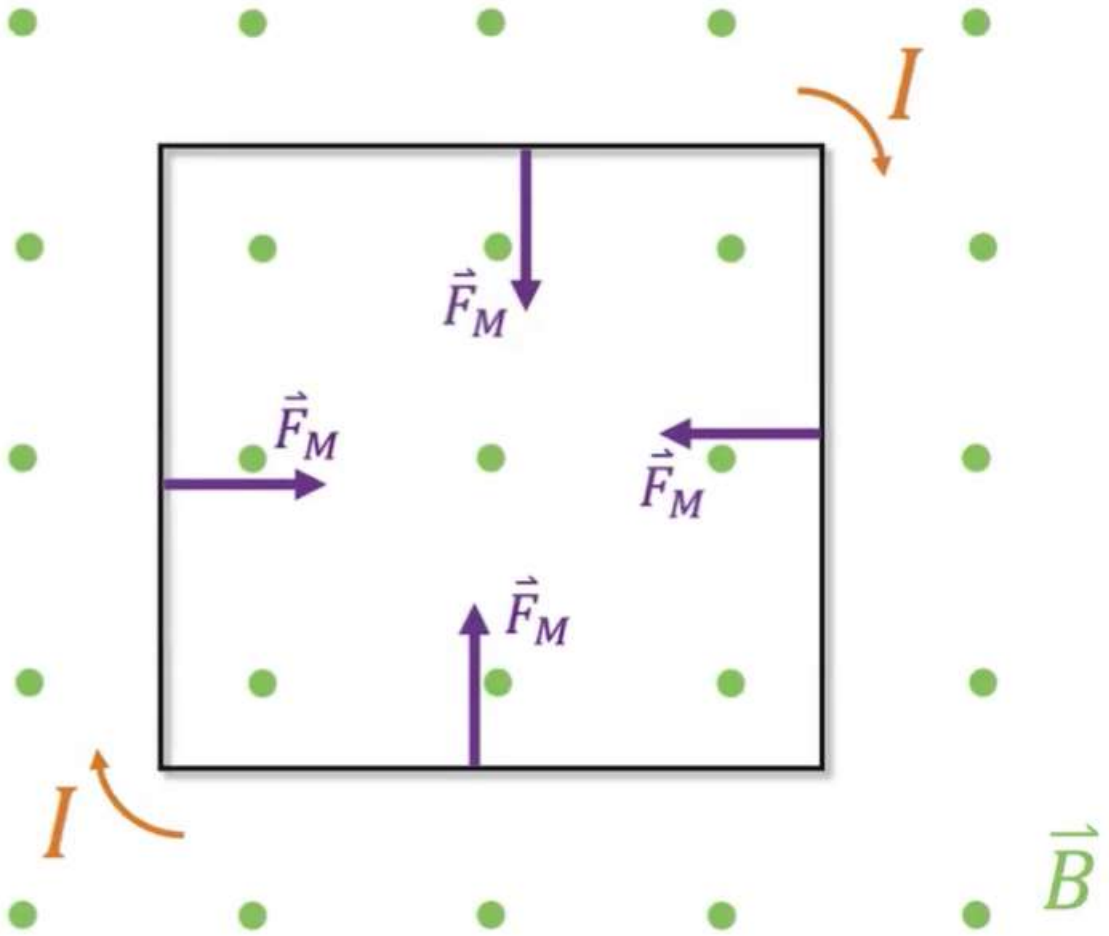
Hint: Use the "Curl" RHR in reverse to help

Decreasing Flux



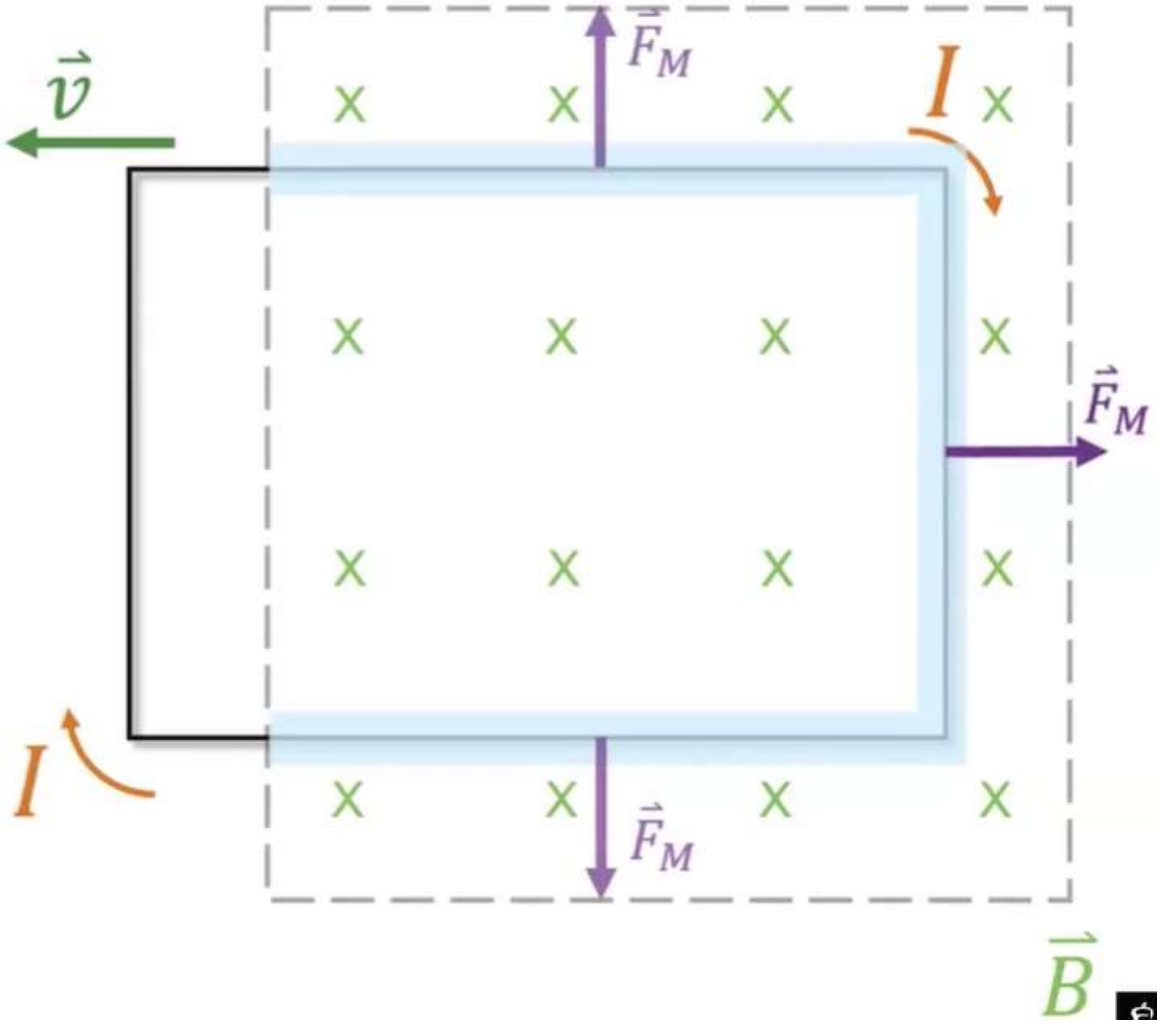
Changing Flux, No Net Magnetic Force

- Magnetic field increases with respect to time, resulting in a changing flux.
- An emf is induced with results in an induced current that opposes the change in flux.
- By individually applying the right hand rule, the magnetic field will exert a force on each section of the loop.
- By symmetry, the force is identical on all sides, resulting in no net magnetic force on the loop.



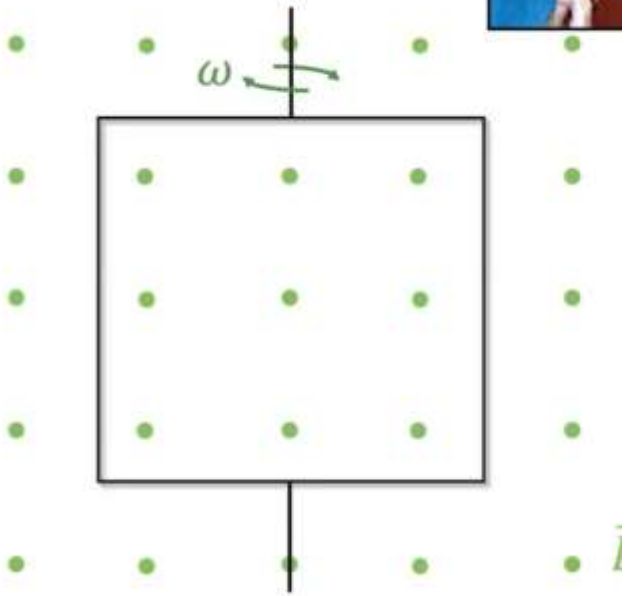
Changing Flux Caused by Translational Motion

- Magnetic field is constant, but defined edges exist.
- A conducting rectangular loop is pulled out of the field.
- This causes a change in flux, inducing an emf, resulting in an induced current.
- By focusing on just the sections of wire *inside* the field, we can use the right-hand rule to determine the direction of the magnetic force.
- By symmetry, the top and bottom forces cancel, leaving a net magnetic force.



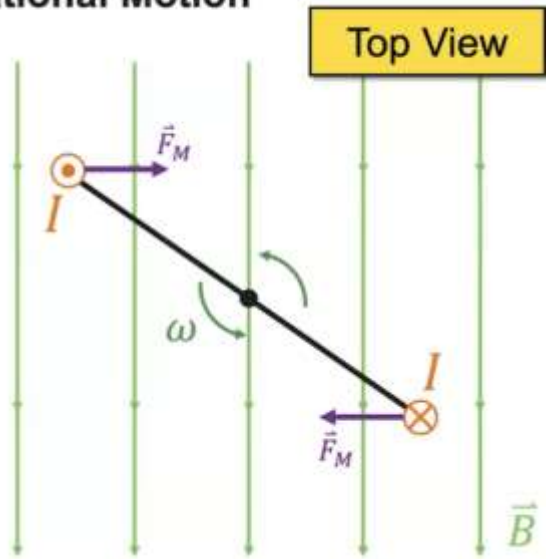
Changing Flux Caused by Rotational Motion

- Square loop is rotated about an axis and entirely inside a constant magnetic field with angular speed ω.
- As the loop rotates, magnetic flux changes, inducing an emf that results in an induced current.
- It is best to visualize this, from a top view, in two ways:
  - Increasing flux
  - Decreasing flux



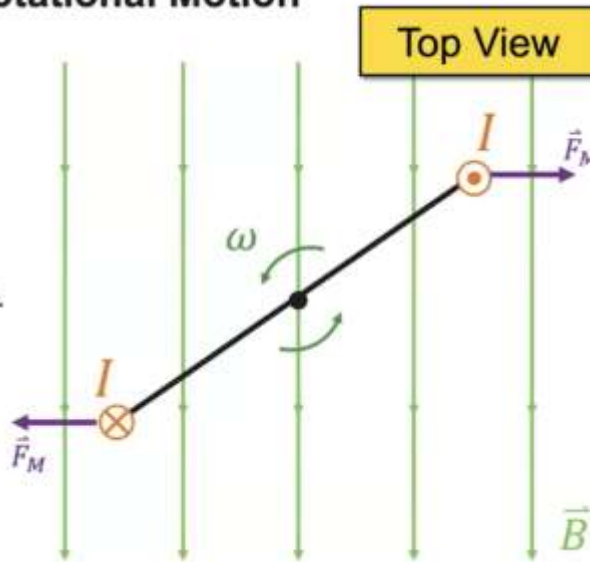
Changing Flux Caused by Rotational Motion

- As the area of the loop rotates and magnetic flux *increases* through the loop, an induced current results from the induced emf.
- By applying the right-hand rule, a magnetic force exists on all four sides.
- By symmetry, the forces on the top and bottom cancel.
- This results in a net torque about the axis of rotation opposite the direction of its angular velocity.



Changing Flux Caused by Rotational Motion

- As the area of the loop rotates and magnetic flux *decreases* through the loop, the induced current changes direction.
- By applying the right-hand rule, a magnetic force exists on all four sides.
- By symmetry, the forces on the top and bottom cancel.
- This results in a net torque about the axis of rotation, again, opposite the direction of its angular velocity.



Takeaways

- Anytime magnetic flux changes through a conducting loop, the induced emf results in an induced current.

- By applying this equation, the magnetic force's magnitude and direction can be determined.

$$\vec{F}_M = \int I(d\vec{l} \times \vec{B})$$

- Only the portion of the conducting loop inside the field will be affected.

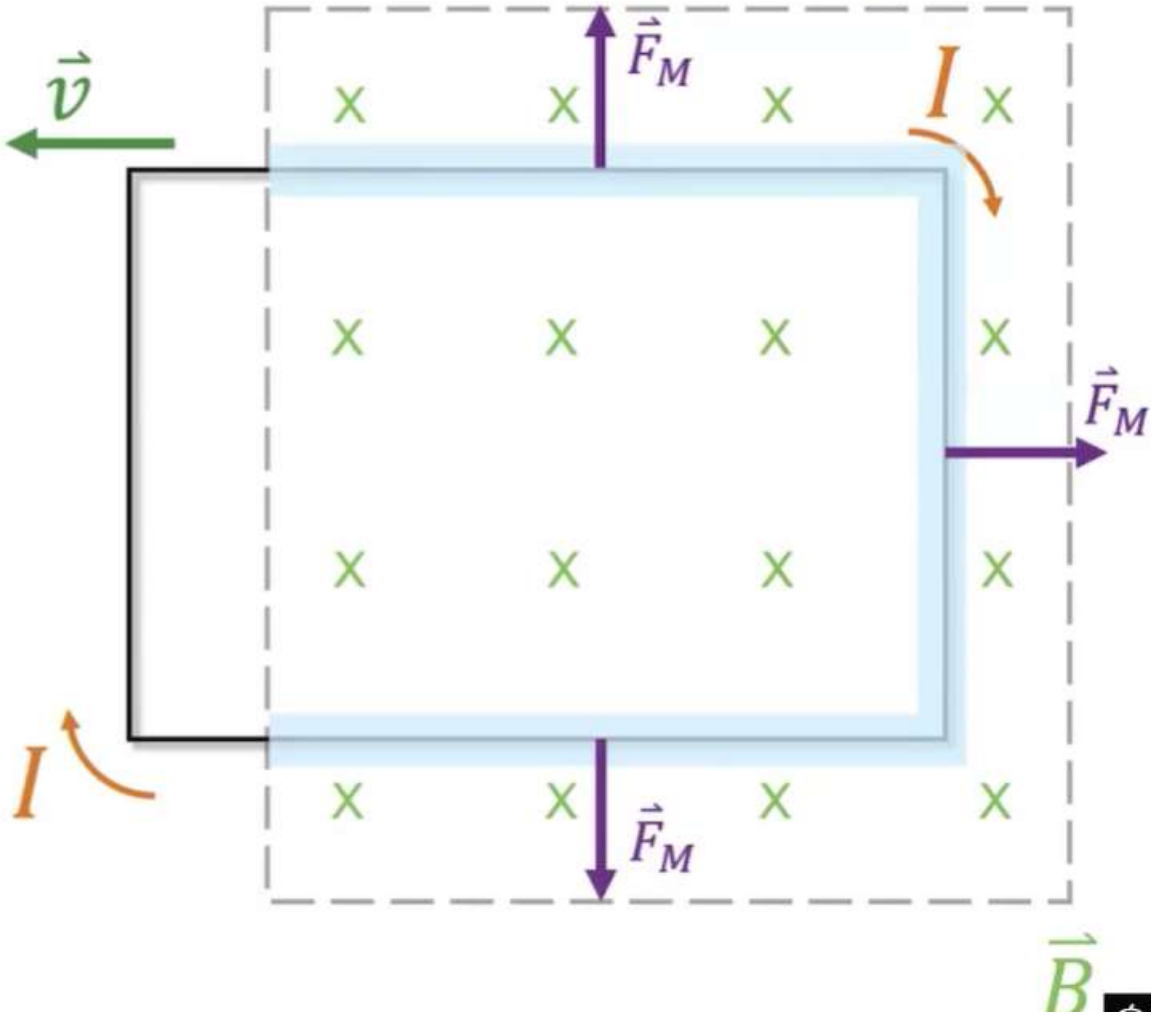
- This may result in a net magnetic force in translational motion or a torque on the loop in rotational motion.

13.3.2

Magnetic Force by an External Field on a moving wire loop carrying an induced current

Changing Flux Caused by Translational Motion

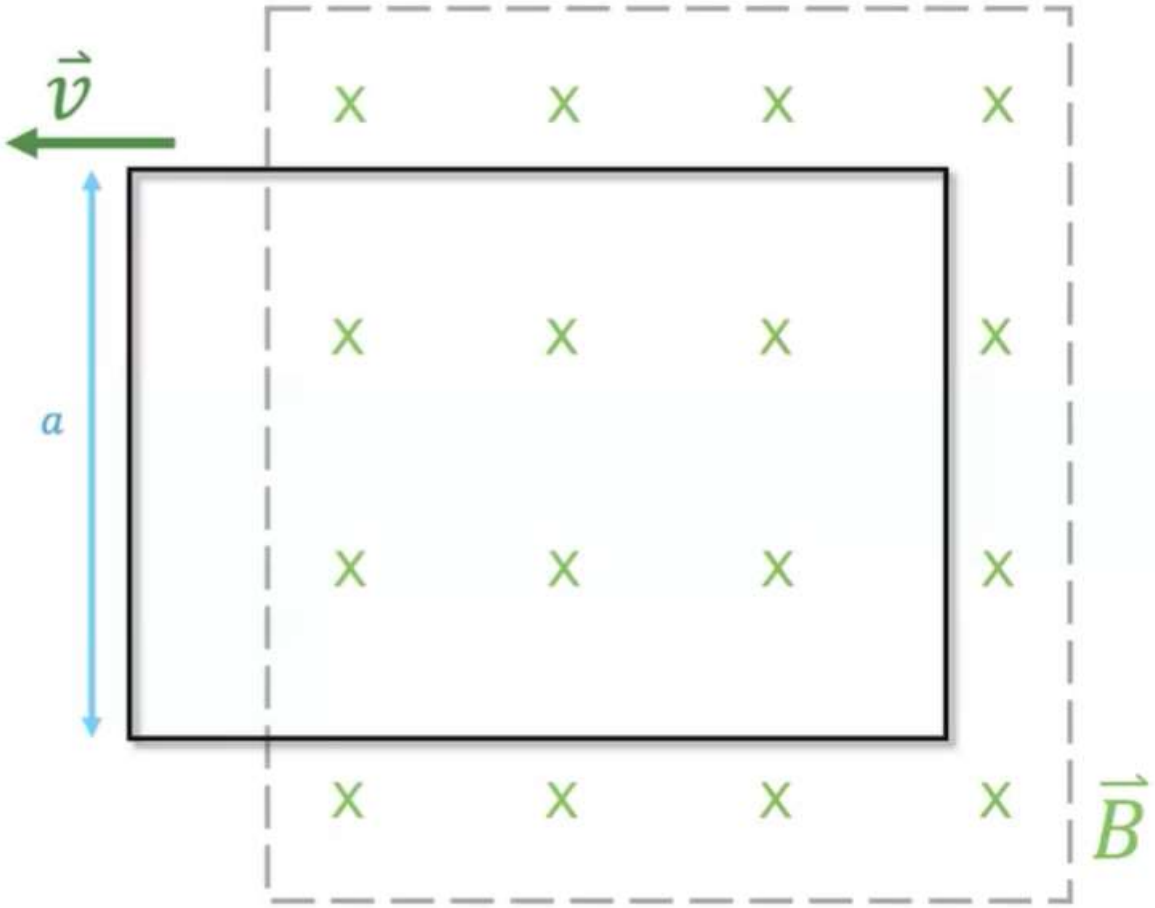
- When a conducting loop moves out of a magnetic field, the magnetic flux inside the loop changes.
- This induces an emf which induces a current in such a direction as to oppose the change.
- By focusing on just the sections of wire *inside* the field, we can use the right-hand rule to determine the direction of the magnetic force.
- By symmetry, the top and bottom forces cancel, leaving a net magnetic force.



Example Problem

What force is necessary to apply to pull the rectangular loop out of the magnetic field while maintaining a constant velocity?

- $a = 0.15\ \mathrm{m}$
- $v = 0.50\ \frac{\mathrm{m}}{\mathrm{s}}$
- $N = 50$
- $R = 0.5\ \Omega$
- $B = 1.25\ \mathrm{T}$



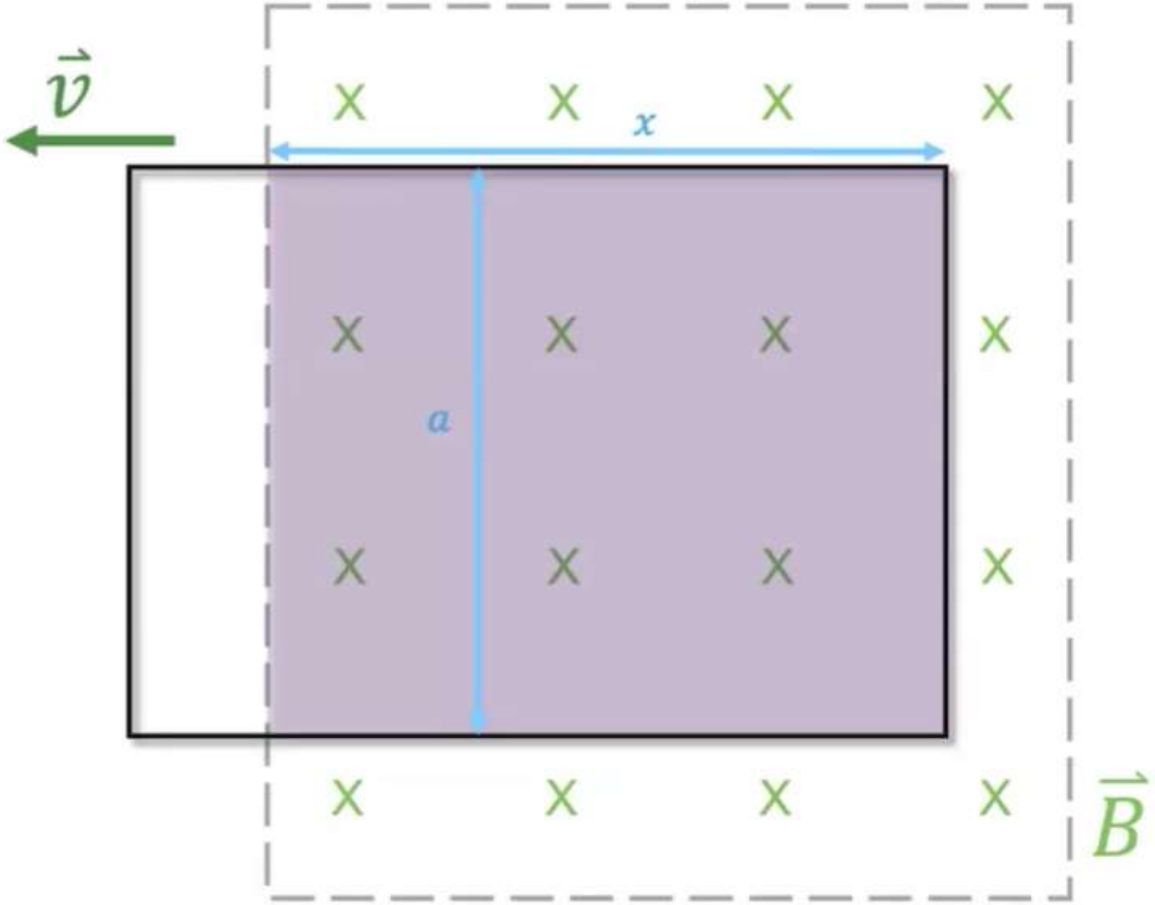
Example Problem

As the loop exists the field, magnetic flux changes, resulting in an induced emf.

$$|\mathcal{E}| = N\left|\frac{d\Phi}{dt}\right|$$

$$\mathcal{E} = N\frac{d(BA)}{dt} = NB\frac{d(A)}{dt}$$

$$\mathcal{E} = NB\frac{d(ax)}{dt} = NBa\frac{dx}{dt}$$



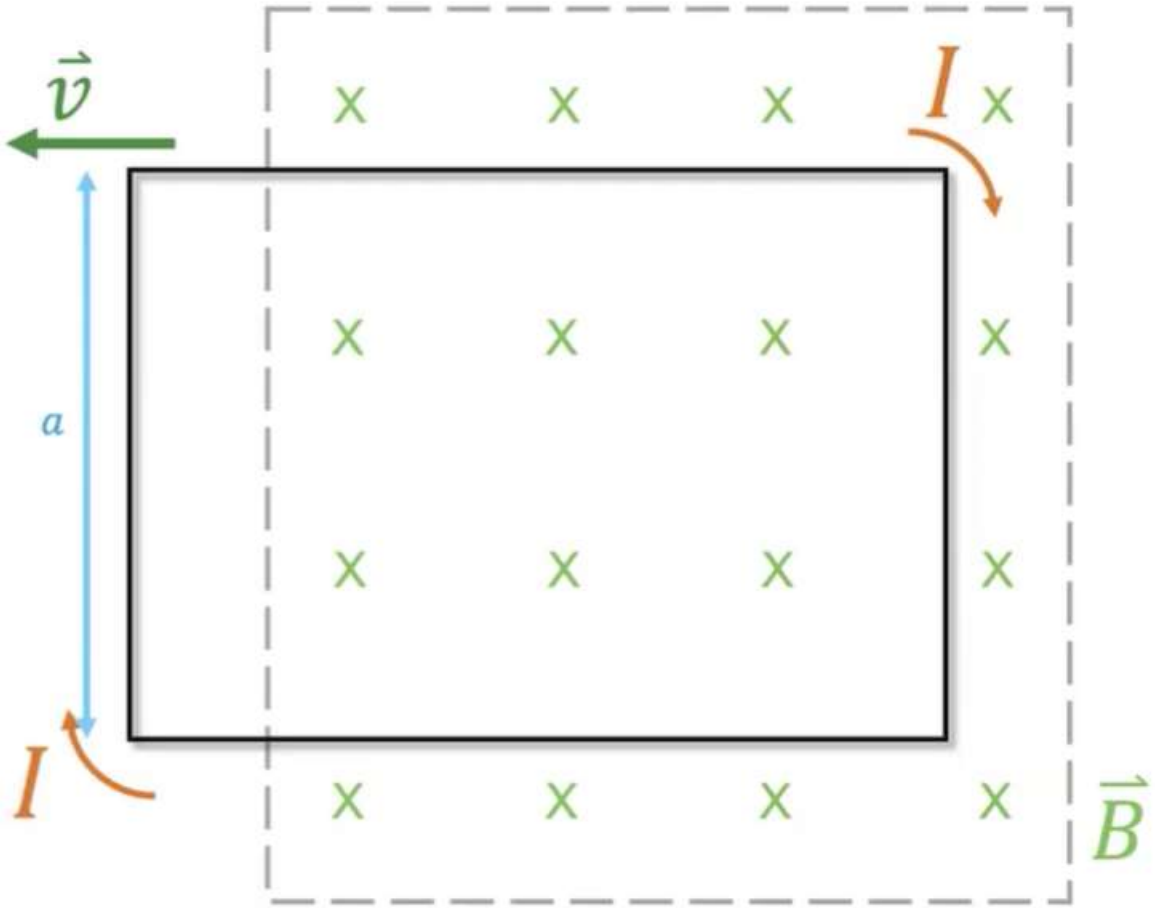
Example Problem

$$\mathcal{E} = NBa\frac{dx}{dt} = NBav$$

Using Ohm's Law, we can relate this emf to current and overall resistance of the loop.

$$I = \frac{\mathcal{E}}{R} = \frac{NBav}{R}$$

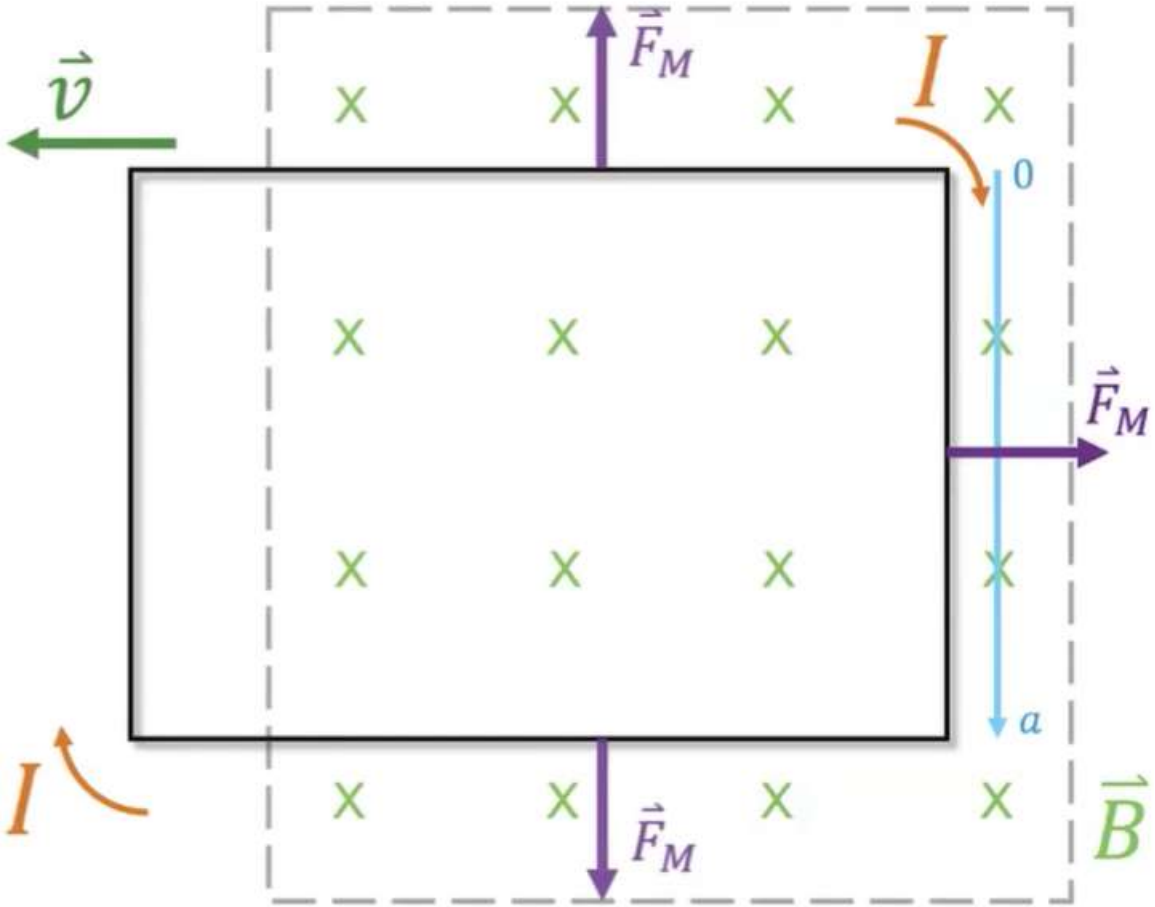
By the right-hand rule, we can determine the direction of this current.



Example Problem

- Using the right-hand rule, we can determine the magnetic force on each section of wire inside the field.
- By symmetry, the top and bottom forces cancel and only the force pictured on the right opposes the motion.
- By setting appropriate limits, we can find the magnitude of this force:

$$\vec{F}_M = \int_0^a I(d\vec{l} \times \vec{B})$$



Example Problem

$$\left|\vec{F}_M\right| = \int_0^a (I)(B)(dl)(\sin\theta)$$

$$F_M = IB\int_0^a dl$$

$$F_M = IB(a-0)$$

$$F_M = IBa$$

By substituting the equation we derived for the induced current we can find the magnitude of the force.

$$I = \frac{NBav}{R}$$

$$F_M = \frac{NBav}{R}Ba$$

$$F_M = \frac{NB^2a^2v}{R}$$

Example Problem

Substituting in given values we can calculate, quantitatively, the value of the magnetic force.

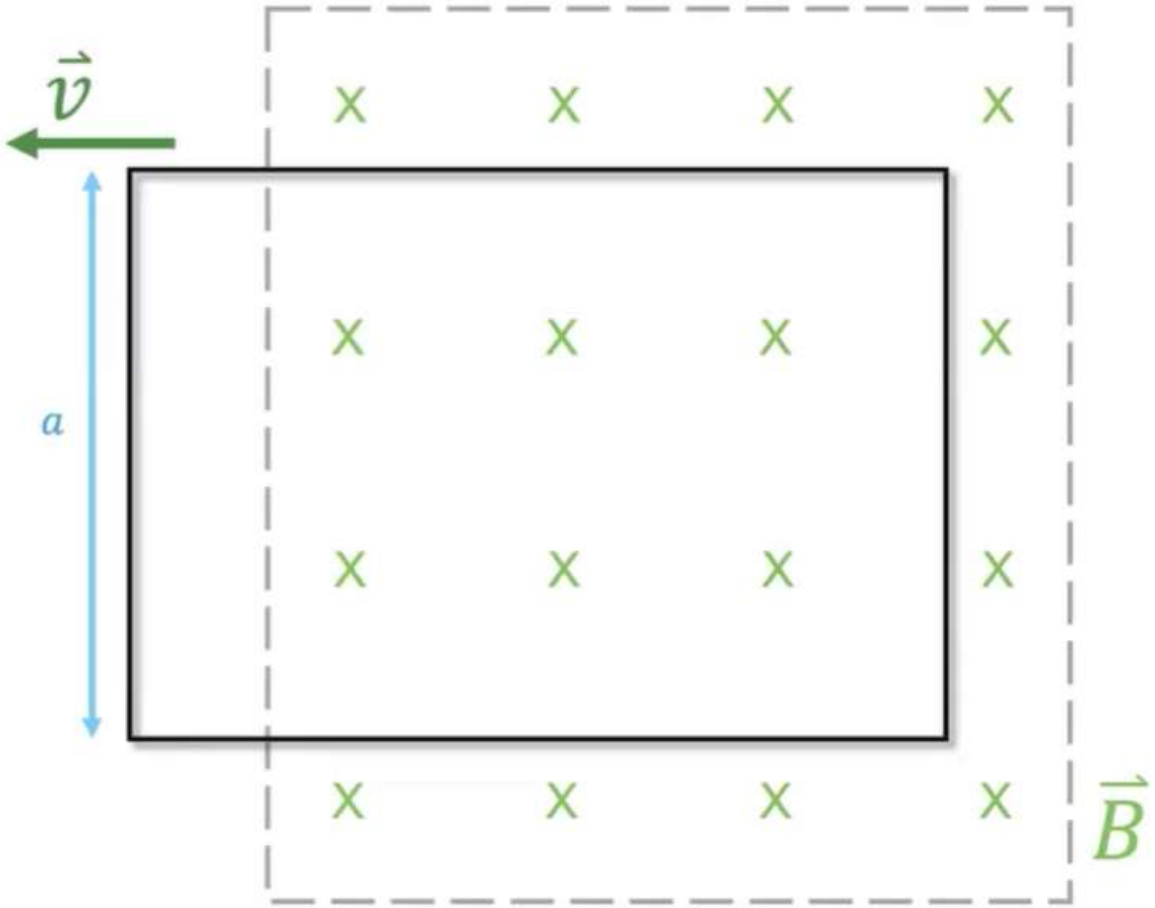
$$F_M = \frac{NB^2a^2v}{R}$$

$$F_M = \frac{50 \cdot (1.25)^2 \cdot (0.15)^2 \cdot (0.5)}{(0.5)}$$

$$F_M = 1.8\ N$$

Example Problem

By drawing a free body diagram, we can clearly see that to maintain a constant velocity, the force that we apply must be equal to the magnetic force induced on the loop.



Takeaways

- Anytime magnetic flux changes through a conducting loop, the induced emf results in an induced current.

- When certain physical characteristic of the the conducting loop are known (size dimensions, resistance) along with other external factors (velocity, magnetic field), the magnetic force that results can be calculated.

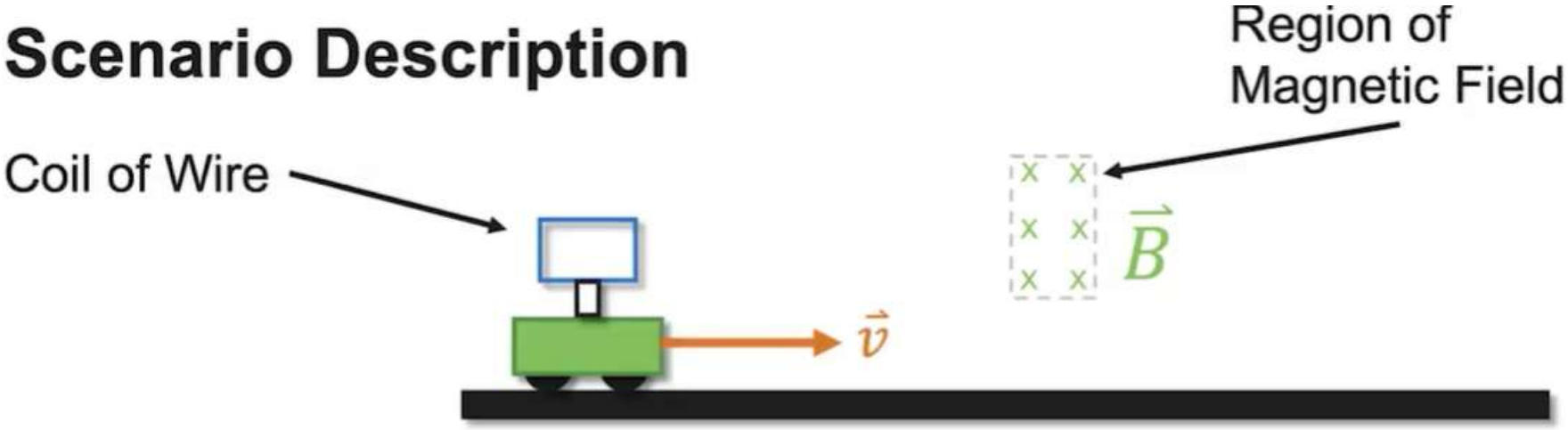
- Once the force is known, kinematic and dynamic principles can be applied to study the resulting motion.

13.3.2

Demonstration of:

Magnetic Force by an External Field on a moving wire loop carrying an induced current

Scenario Description

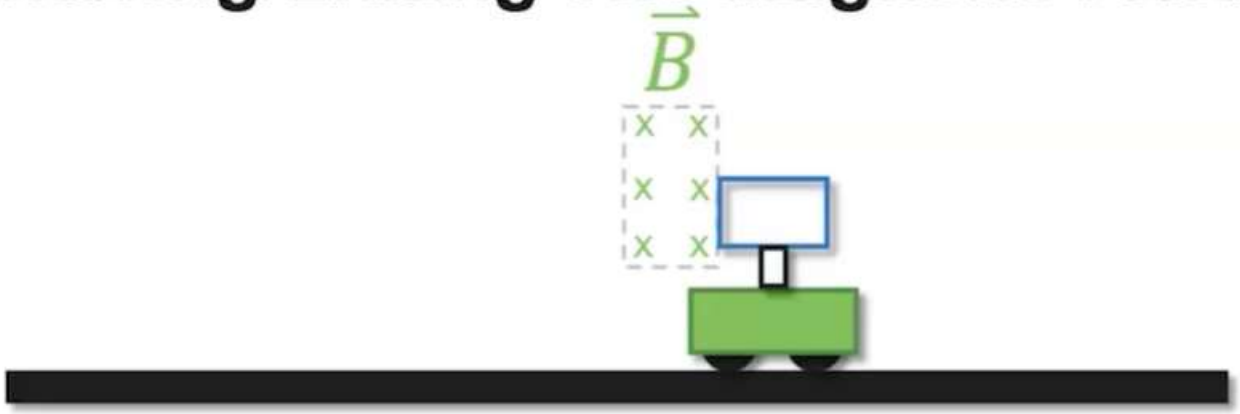


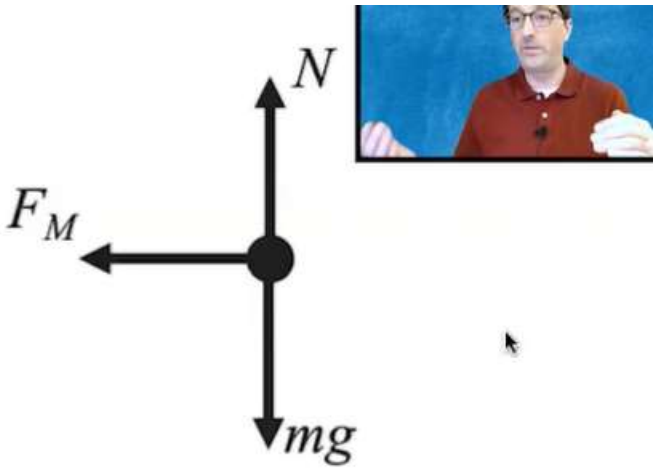
- A cart with a coil of wire attached to the top of travels along a frictionless track.
- As the cart travels the coil of wire will pass through a region with a constant magnetic field present.
- By application of the concepts of magnetic forces that result on induced forces in conducting loops, we should be able to **predict the shape of the cart's velocity vs. time graph**.
- We will predict this and then set this up in a lab setting to verify.

Three Analysis Regions

- For simplicity, the coil of wire has the same width as the region containing the magnetic field.
- Three regions exist to analyze:
  - Before the coil enters the magnetic field ← Constant Velocity
  - While the coil is entering/exiting
  - After the coil exits the field ← Constant Velocity

Entering/Exiting The Magnetic Field



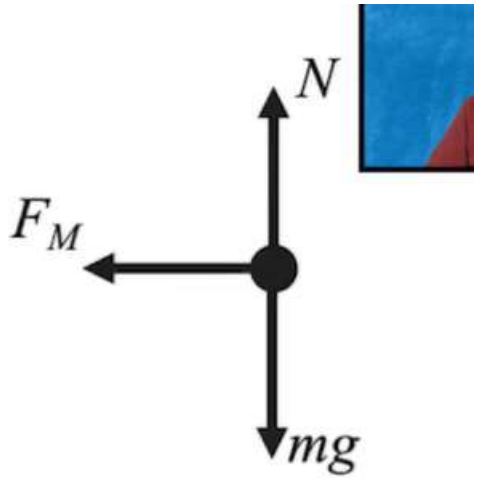


- From the previous video, we determined the strength of the magnetic force induced to be:

$$F_M = \boxed{\frac{NB^2a^2}{R}}v \qquad F = -\boxed{k}v$$

- In this example, the magnetic force acts just like a resistive force discussed in the Mechanics course: a velocity dependent force acting opposite the direction of velocity.

Making a prediction



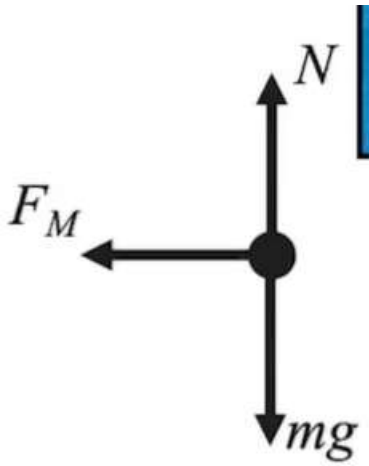
$$F_M = \frac{NB^2a^2}{R}v$$

$$\sum F = ma = -\frac{NB^2a^2}{R}v \qquad \text{let } k = \frac{NB^2a^2}{R}$$

$$ma = -kv$$

$$m\frac{dv}{dt} = -kv \quad \rightarrow \quad \frac{dv}{v} = -\frac{k}{m}dt$$

Making a prediction



$$\frac{dv}{v} = -\frac{k}{m}dt$$

$$\int_{v_i}^{v(t)} \frac{dv}{v} = -\frac{k}{m}\int_0^t dt$$

$$\ln\big(v(t)\big) - \ln(v_i) = -\frac{k}{m}(t-0)$$

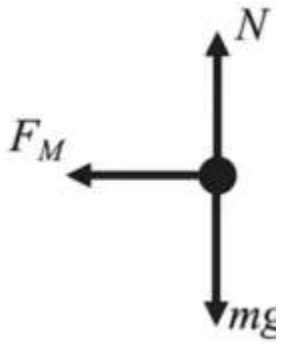
$$\ln\left(\frac{v(t)}{v_i}\right) = -\frac{k}{m}t$$

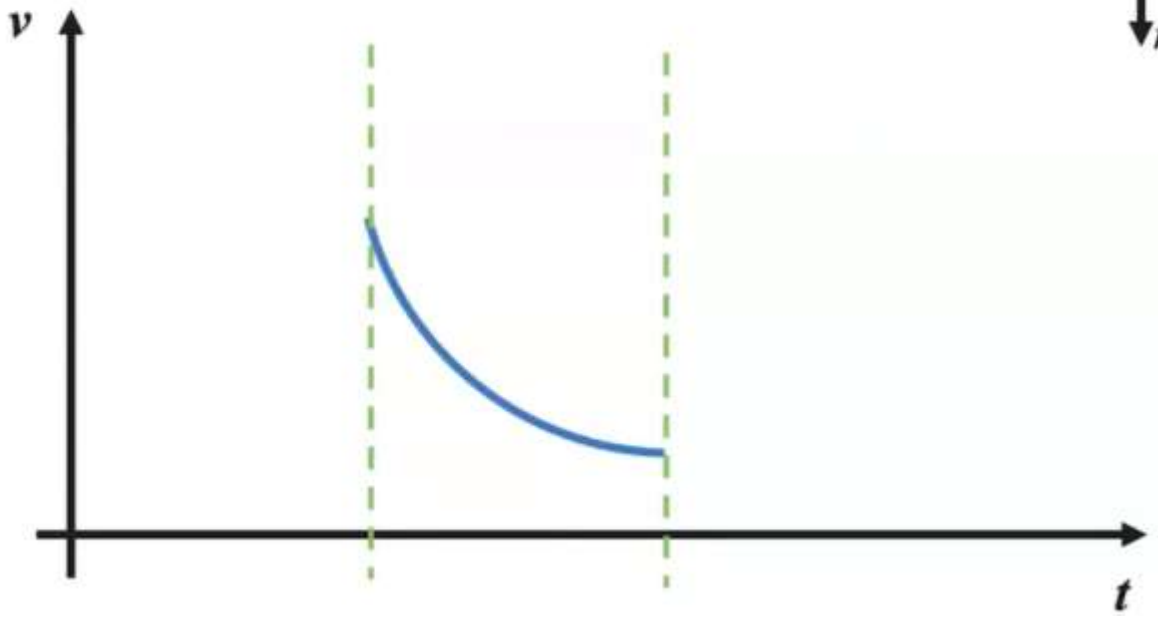
$$\frac{v(t)}{v_i} = e^{-\frac{k}{m}t}$$

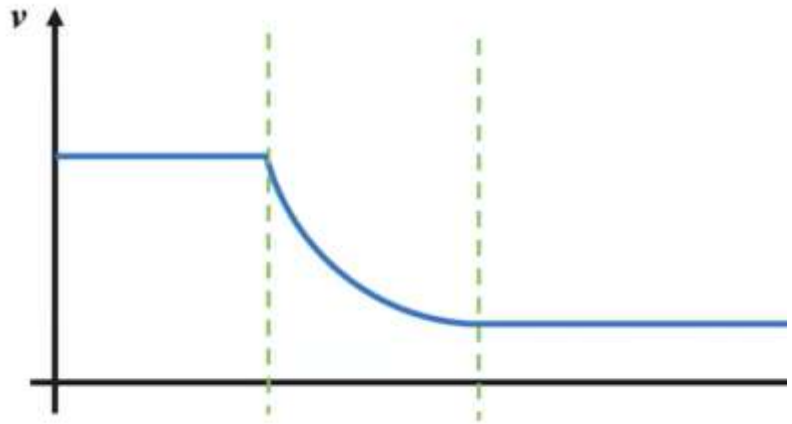
$$v(t) = v_i\, e^{-\frac{k}{m}t}$$

Graphing the Prediction

$$v(t) = v_i\, e^{-\frac{k}{m}t}$$



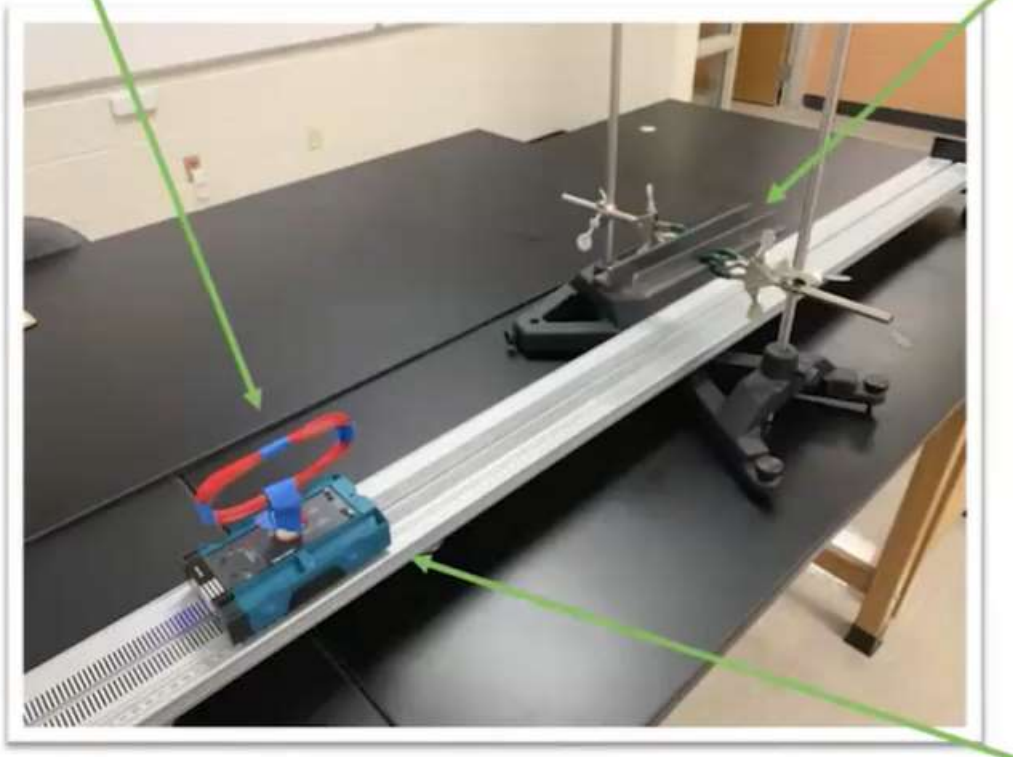




Testing the Prediction

Region of Magnetic Field

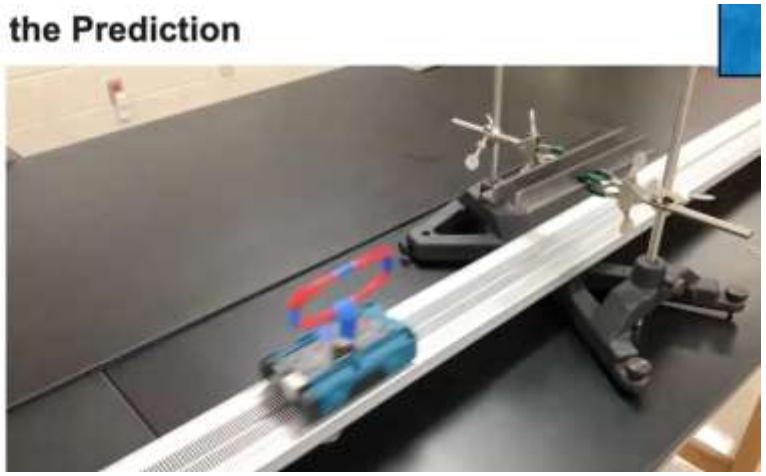
Coil of Wire



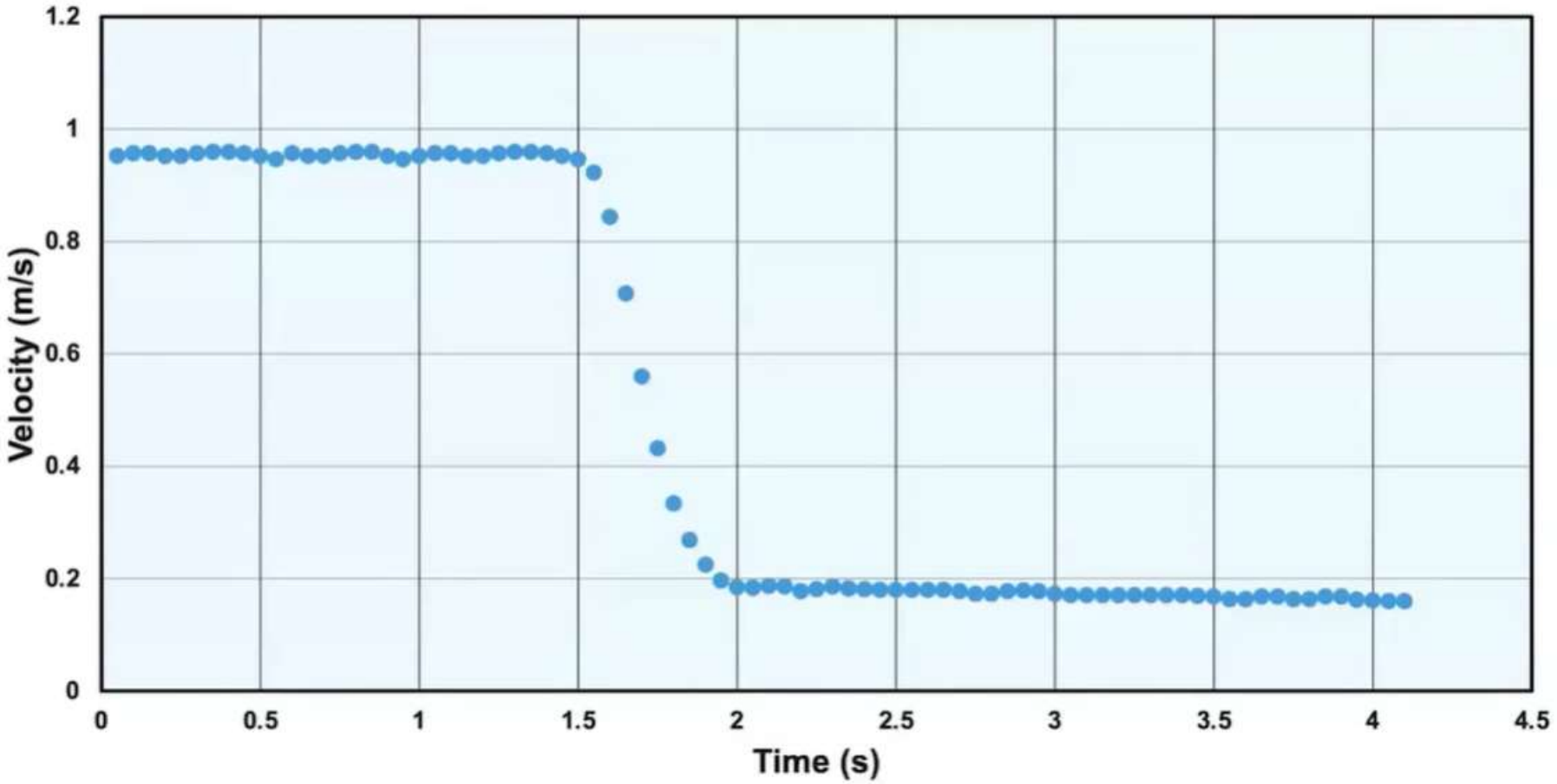
- A motion encoder cart and sensor is used to accurately track the cart's velocity as it rolls down the track.
- A coil of wire is attached to the top of the cart.
- A magnetic field is set up using strong, neodymium magnets set in the brackets.

Motion Encoder Cart

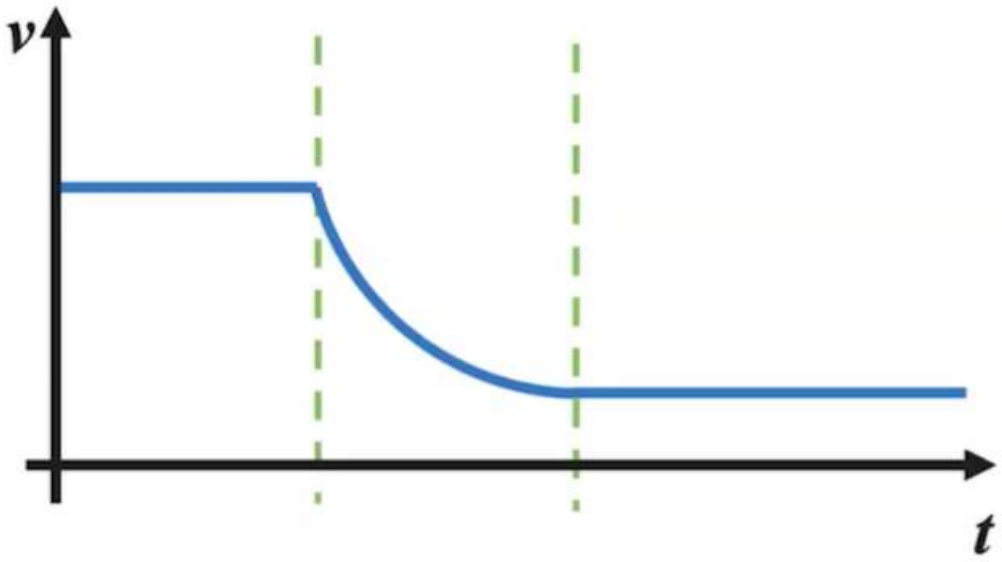
Testing the Prediction

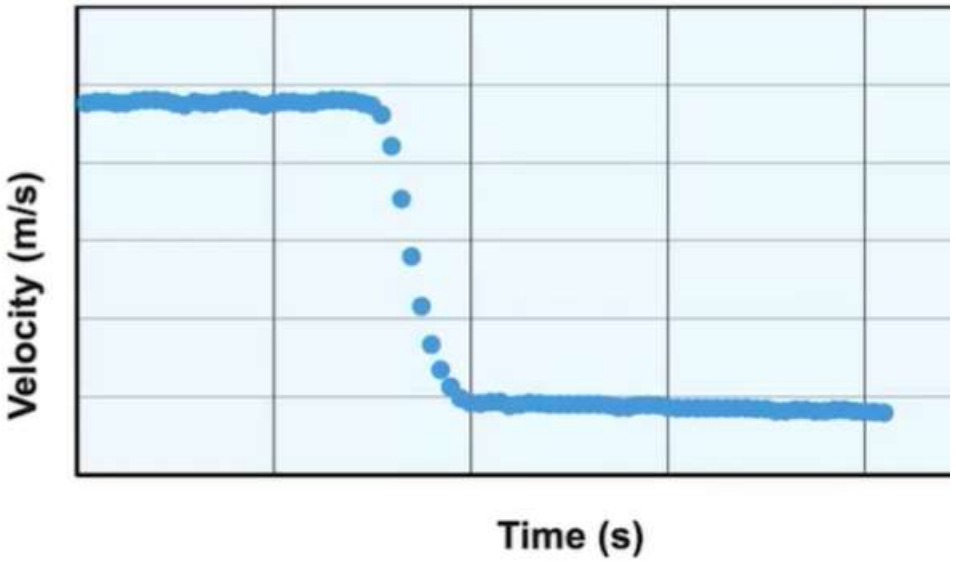


Looking at the Data



Comparing Results





Takeaways

- This prediction and test should act as confirmation of
  - A changing magnetic flux through a conducting coil induces an emf that results in an induced current.
  - The magnetic field exerts a force on wires as this induced current flows.
- By applying Newton's second law to the forces acting on the cart-loop system, we were able to create a prediction of a velocity vs. time graph that was confirmed through experimentation in a lab setting.